

ACML100
Major, 2024
Marks: 40

(Prof. Shashank Deep, Prof. Nidhi Jain, Prof. Subrata Kundu)

(Important Note: There will be step wise marking. Make sure that every step is written in answer sheet. Skipping of any step will result into deduction of mark.

- (1) Evaluate the following commutator: (2)

$$[\hat{H}, \hat{p}_x] \quad \text{where} \quad \hat{H} = \frac{\hat{p}_x^2}{2m} + V(x) \quad \text{and} \quad V(x) = \frac{1}{2} k_f x^2 \quad \text{and} \quad \underline{k_f \text{ is a constant}}$$

- (2) The momentum operator is proportional to d/dx. Which of the followings are eigen functions of d/dx? (i) e^{ikx} (ii) $e^{\alpha x^2}$ (2)

- (3) A long carbon nanotube of 100 nm length can be modelled as a one-dimensional structure and its electron described by particle-in-a-box wavefunction. What will be wavelength of the light emitted on transmission of electron from $n=2$ to $n=1$. (2)

- (4) The wavefunctions for the radial and angular parts of a certain orbital of a hydrogen-like atom are:

$$R_{nl} = \left(\frac{Z}{a_0}\right)^{3/2} \left(\frac{1}{2430}\right)^{1/2} \rho^2 \exp(-\rho/2); \quad \rho = \left(\frac{2Z}{3a_0}\right) r;$$

$$Y_{l,m_l} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta \exp(-2i\phi)$$

$$\psi = R_{nl} Y_{l,m_l}$$

- (a) What are the n , l , and m_l values of this orbital? (1)

- (b) Find the most probable distance (radial) for this orbital. (1)

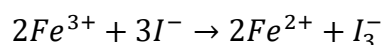
(Given: Volume element (in spherical polar coordinates): $d\tau = r^2 \sin \theta dr d\theta d\phi$ and

$$\int_0^\pi \int_0^{2\pi} Y_{l,m_l}^* Y_{l,m_l} \sin \theta d\theta d\phi = 1)$$

- (c) Calculate the expectation value $\langle r \rangle$ for an electron having this wavefunction. (1)

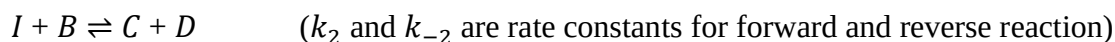
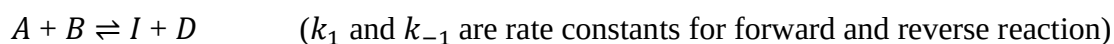
$$\text{(Given, } \int_0^\infty x^n \exp(-ax) dx = \frac{n!}{a^{n+1}})$$

- (5) Propose reaction steps consistent with the rate law for the reaction between Fe^{3+} and I^- . Also write the rules used for deriving the mechanism (3)



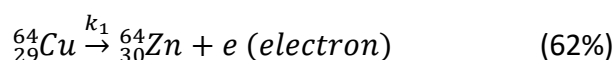
$$-\frac{d[Fe^{3+}]}{dt} = \frac{k[Fe^{3+}]^2[I^-]^2}{[Fe^{3+}] + k'[Fe^{2+}]}$$

(6) A reaction $A + 2B \rightleftharpoons C + 2D$ proceeds through the mechanism:



Applying steady state approximation to I , express $-\frac{d[A]}{dt}$ in terms of rate constant and concentration of reactant and product. **(3)**

(7) Radioactive nuclei ${}^{64}_{29}\text{Cu}$ decay frequently occurs simultaneously by two separate first-order paths **(2)**



The decay of ${}^{64}_{29}\text{Cu}$ occurs exponentially with a single half-life of 12.8 h. Calculate the rate constant of two paths (k_1 and k_2)

Questions 8-11 choose the correct answer

(1.5 x 4 = 6)

(8) If Δ_0 is the octahedral splitting energy and P is the pairing energy, then crystal field stabilization energy (CFSE) of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ is

- (a) $-0.8\Delta_0 + 2P$ (b) $-0.8\Delta_0 + 1P$ (c) $-0.8\Delta_0$ (d) $-1.8\Delta_0 + 1P$

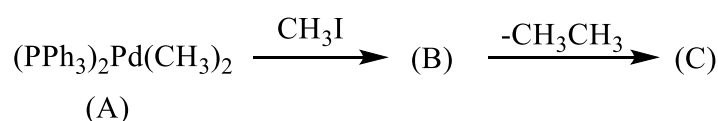
(9) The complex $[\text{Fe}(\text{Phen})_2(\text{NCS})_2]$ (phen = neutral nitrogen based chelating ligand) shows spin cross-over behavior ($T_{1/2} = 175 \text{ K}$). CFSE (excluding pairing energy) and μ_{eff} at 250 and 100 K respectively are

- (a) $-0.4 \Delta_0$, 4.90 BM and $-2.4 \Delta_0$, 0.00 BM (b) $-2.4 \Delta_0$, 2.90 BM and $-0.4 \Delta_0$, 1.77 BM
(c) $-2.4 \Delta_0$, 0.00 BM and $-0.4 \Delta_0$, 4.90 BM (d) $-1.2 \Delta_0$, 4.90 BM and $-2.4 \Delta_0$, 0.00 BM

(10) Among the complexes (A) $[\text{Mn}(\text{H}_2\text{O})_6]\text{Cl}_2$, (B) $\text{K}_4[\text{Cr}(\text{CN})_6]$, and (C) $\text{K}_3[\text{Co}(\text{CN})_6]$ Jahn-Teller distortion is expected in

- (a) only in A (b) A, B, C (c) only in B (d) B and C

(11) Consider the following steps of a catalytic reaction and state the correct oxidation state of the metal complexes A, B and C respectively



- (a) 2, 0, 2 (b) 2, 4, 2 (c) 2, 4, 4 (d) 2, 3, 2

Questions 12-17: Comment the statements as true or false

(6 x 1 = 6)

(12) KMnO_4 exhibits charge transfer transitions; therefore, it has lower intensity and a lower molar extinction coefficient.

(13) Among the following metal complexes— $\text{K}_3[\text{Mn}(\text{CN})_6]$ (**A**), $\text{K}_3[\text{Fe}(\text{CN})_6]$ (**B**), and $\text{K}_3[\text{Co}(\text{CN})_6]$ (**C**)—the central metal ion with the lowest ionic radius is found in complex **B**.

(14) According to VSEPR theory, the structure of XeF_4 is square planar, and the structure of XeF_2 is linear.

(15) The total number of metal-metal bonds found in $\text{Co}_4(\text{CO})_{12}$ is six

(16) Mn_3O_4 has inverse spinal structure.

(17) $\text{Ni}(\text{CO})_4$, $[\text{Co}(\text{CO})_4]^-$, and $[\text{Fe}(\text{CO})_4]^{2-}$ follow the 18-electron rule; therefore, their carbonyl stretching frequencies (ν_{CO}) in the IR spectrum will be the same.

Questions 18-21: Give descriptive answer

(4 x 2 = 8)

(18) Explain the Mond process used for the purification of nickel.

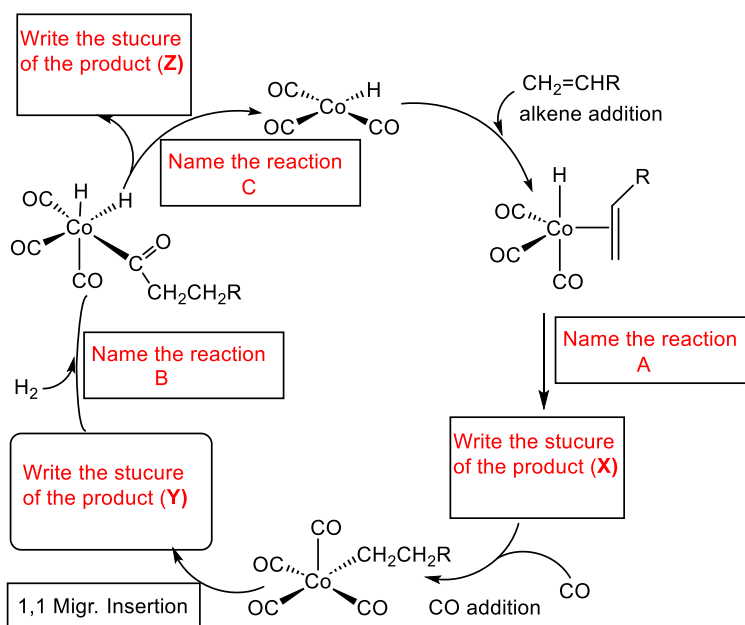
(19) The complex $(\text{PPh}_3)_3\text{Rh}(\text{Me})$ in the presence of D_2 forms methane and $(\text{PPh}_3)_2\text{Rh}(\text{D})(\text{PPh}_2\text{C}_6\text{H}_4\text{D})$. Write the two important steps involve in this reaction.

(20) Among $[\text{Fe}(\text{SCN})_3(\text{H}_2\text{O})_3]$, $[\text{NiCl}_4]^{2-}$, and $[\text{Mn}(\text{H}_2\text{O})_6]\text{Cl}_2$, which will exhibit low-intensity color and which will exhibit high-intensity color, and why?

(21) (a) Write the geometry and structure of $[\text{ClO}_2]^-$ and comment on the deviation of the O-Cl-O bond angle from the ideal geometry. (b) Write the structure of $[\text{IO}_2\text{F}_2]^-$.

Question 22: Find out the missing steps [3]

A catalytic cycle for a hydroformylation reaction is shown below. In this cycle, the names of the reactions (A, B and C) and the structures of the products (X, Y and Z) in three steps are missing. Provide the names of the missing reactions (A, B and C) and the structures of the products (X, Y and Z).



.....End.....

Group	1	2	3	4	5	6	7	8	9	10	11	1	17	18					
Period	↓																		
1	1 H													2 He					
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne	
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
6	55 Cs	56 Ba	*	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	**	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			*	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			**	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		